Utilizing CLF and CBF on Adaptive Cruise Control

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Section 1

1. Motivation
2. Introduction
3. ACC Problem Setup
4. Experiment
5. Conclusion

Adaptive Cruise Control

- Adaptive Cruise Control is a Safety Critical problem
- Two conflicting objectives
  - Maintain desired speed (performance)
  - Keep an appropriate distance (safety)

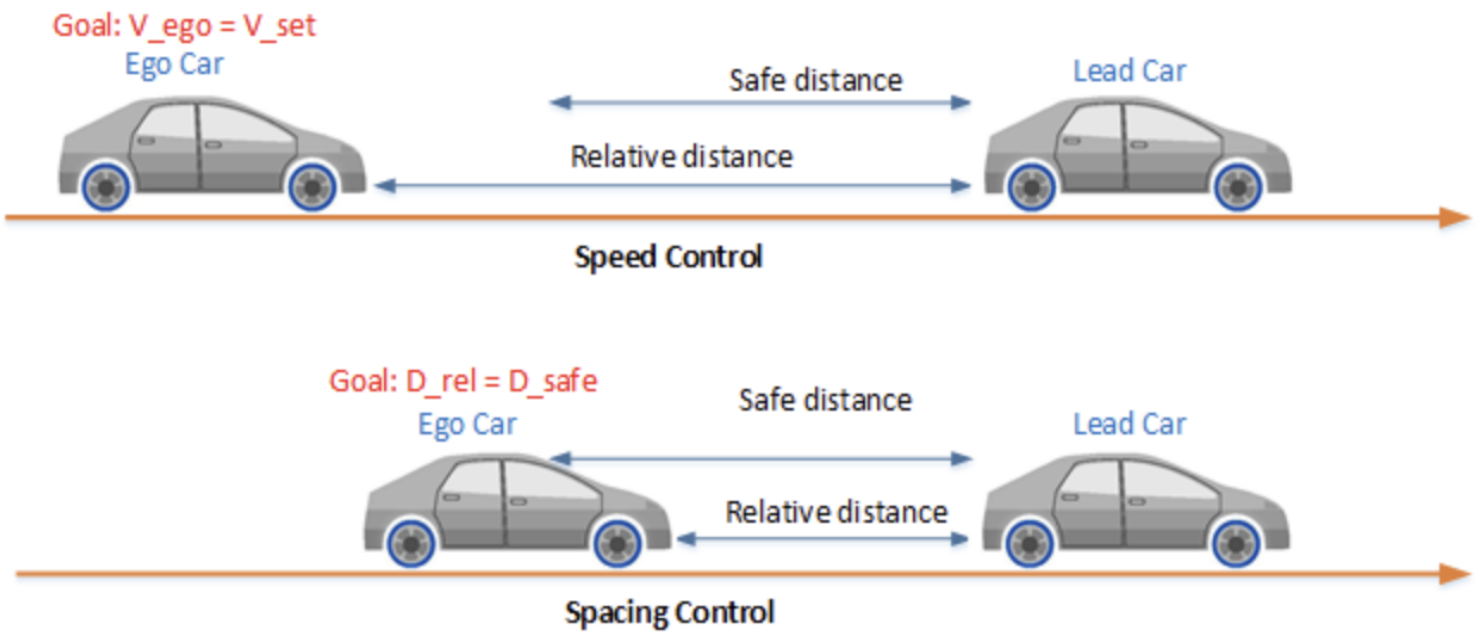

Figure: Illustration of the ACC (Source: Mathworks)

Previous Work

- PID Controller Framework
  - **Plant (Transfer Function):**

$$V(s) = P(s)U(s), \quad P(s) = \frac{1}{ms + b}$$

  - **Error:**

$$E(s) = V_{\text{ref}}(s) - V(s)$$

  - **Controller:**

$$U(s) = C(s)E(s)$$

- Model Predictive Control

At each time step, MPC solves:

$$\begin{aligned} \min_{u(0),\ldots,u(N-1)} \quad & \sum_{k=0}^{N-1} \ell(x(k), u(k)) + V_f(x(N)) \\ \text{s.t.} \quad & x(k+1) = f(x(k), u(k)) \\ & x(k) \in \mathcal{X}, \quad u(k) \in \mathcal{U} \end{aligned}$$

State constraint set

$$\mathcal{X} = \left\{ x = \begin{bmatrix} \delta_d \\ \delta_v \\ v_h \end{bmatrix} \middle| \begin{array}{l} 0 \leq \delta_d \leq 2 \\ 0 \leq \delta_v \leq 2.5 \\ 15 \leq v_h \leq 40 \end{array} \right\}$$

Input constraint set

$$\mathcal{U} = \{u \mid -3 \leq u \leq 5\}$$

Proposal of Paper

- What this paper proposes
  - CLF (Control Lyapunov Function) : Drive System toward goal
  - CBF (Control Barrier Function): Enforce safety constraints

Section 2

1. Motivation
2. **Introduction**
3. ACC Problem Setup
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Lyapunov Theory

In this section, talk about how is the stability of the system being analyzed. Suppose the linear system is of the form

$$\dot{x} = f(x) \tag{1}$$

where f is locally lipschitz. We try to find a positive function that decrease as the system evolve. We define a Lyapunov function

$$\begin{aligned} & V(x) \\ \text{s.t. } & V(x_e) = 0, \quad V(x) > 0 \text{ for } x \neq x_e, \\ & \dot{V}(x) = \frac{\partial V}{\partial x} f(x) < 0 \text{ for } x \neq x_e \end{aligned} \tag{2}$$

Barrier Function

In this section, we talk about how to construct a barrier function $B(x)$ that achieves forward invariance, which is when we When does a set of state guaranteed to be safe forever under a dynamical system? Goal: Construct a barrier function $B(x)$ achieves forward invariance

- Safe Set:

$$\mathcal{C} = \{x \in \mathbb{R}^n : h(x) \geq 0\}, \tag{3}$$

$$\partial\mathcal{C} = \{x \in \mathbb{R}^n : h(x) = 0\}, \tag{4}$$

$$\mathsf{Int}(\mathcal{C}) = \{x \in \mathbb{R}^n : h(x) > 0\}. \tag{5}$$

- the barrier function candidate

$$B(x) = -\log(\frac{h(x)}{1 + h(x)}) \tag{6}$$

Property

$$\inf_{x \in \mathsf{Int}(\mathcal{C})} B(x) > 0, \qquad \lim_{x \to \partial\mathcal{C}} B(x) = \infty. \tag{7}$$

The condition imposed on $\dot{B}$ is

$$\dot{B} \leq \frac{\gamma}{B} \tag{8}$$

Lie Derivative Notation

Suppose we have a affine control system

$$\dot{x} = f(x) + g(x)u \tag{9}$$

and a barrier function

$$B(x) \tag{10}$$

We want to know that how does $B(x)$ change over time as the state evolves.

$$\dot{B}(x) = \nabla B(x)\dot{x} = \nabla B(x) \cdot (f(x) + g(x)u) \tag{11}$$

We define the following, which is called the lie derivatives.

$$L_f B(x) := \nabla B(x) \cdot f(x) \tag{12}$$

$$L_g B(x) := \nabla B(x) \cdot g(x) \tag{13}$$

Therefore, we have

$$\dot{B}(x) = L_f B(x) + L_g B(x)u \tag{14}$$

Control Barrier Function

Suppose we have a affine control system

$$\dot{x} = f(x) + g(x)u \tag{15}$$

with f and g locally Lipschitz, $x \in \mathbb{R}^m$ and $u \in U \subset \mathbb{R}^m$. Design a controller to ensure containment in $\mathcal{C}$

Definition 2: Let $\mathcal{C} \subset \mathbb{R}^n$ be defined by (2)-(4) for a continuously differentiable function $h : \mathbb{R}^n \to \mathbb{R}$, then a function $B : \mathcal{C} \to \mathbb{R}$ is a **control barrier function (CBF)** if there exist class $\mathcal{K}$ functions α_1, α_2 and $0 < \gamma$ such that

$$\frac{1}{\alpha_1(h(x))} \leq B(x) \leq \frac{1}{\alpha_2(h(x))} \tag{16}$$

$$\inf_{u \in U} \left[L_f B(x) + L_g B(x) u - \frac{\gamma}{B(x)} \right] \leq 0 \tag{17}$$

for all $x \in \text{Int}(\mathcal{C})$.

CLF and ES-CLF

Goal: Choose a control input u such that $V(x)$ decreases over time.
Let $V(x) : \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable function. If there exists a constant $c > 0$ such that:

1. **Bounded sublevel set**

$$\Omega_c := \{x \in \mathbb{R}^n : V(x) \leq c\}$$

2. **Positive definiteness**

$$V(x) > 0 \text{ for all } x \neq x_e, \qquad V(x_e) = 0$$

3. **Control-dependent decrease condition**

$$\min_{u \in U} \dot{V}(x, u) < 0 \quad \forall x \in \Omega_c \setminus \{x_e\}$$

Then $V(x)$ is a (local) **Control Lyapunov Function (CLF)**
If there exists some constant $\lambda > 0$ such that

$$\min_{u \in U} \dot{V}(x, u) + \lambda V(x) \leq 0,$$

V(x) is **Exponentially Stabilizing-Control Lyapunov Function (ES-CLF)**.

CLF-CBF-QP

Given a ES-CLF, V, and a CBF, B, they can be combined into a single controller through the use of a QP of the form

$$
\begin{aligned}
u^*(x,z) = \quad & \underset{\mathbf{u}=\begin{bmatrix} u \\ \delta \end{bmatrix} \in \mathbb{R}^{m+1}}{argmin} && \frac{1}{2}\mathbf{u}^T H(x,z)\mathbf{u} + F(x,z)^T\mathbf{u} && \text{(CLF-CBF QP)} \\
& \text{s.t.} && \psi_0(x,z) + \psi_1^T(x,z)u \leq \delta && \text{(CLF)} \\
& && L_f B(x,z) + L_g B(x,z)u \leq \frac{\gamma}{B(x,z)} && \text{(CBF)}
\end{aligned}
$$

Section 3

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ACC Problem Setup

Let $x = (x_1, x_2)$ with x_1 the position of the vehicle, and x_2(or v) $= \dot{x_1}$ the velocity. Define $z = D$ be the distance between the vehicle, and the second vehicle traveling at constant velocity v_0.
The dynamics of the system

$$\dot{x} = \underbrace{\begin{bmatrix} x_2 \\ -\frac{1}{m}F_r \end{bmatrix}}_{f(x,z)} + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}}_{g(x,z)} u \tag{18}$$

$$\dot{z} = \underbrace{v_0 - x_2}_{q(x,z)},$$

where $u = F_w$ is the control input, and $F_r = f_0 + f_1 x_2 + f_2 x_2^2$ denotes the aerodynamic drag.

Constraint I: Hard Constraint as CBF

keep a safe distance from the car in front of you

$$D \geq \frac{v}{2} \tag{19}$$

Expression:

$$L_f B(x,z) + L_g B(x,z) u - \frac{\gamma}{B(x,z)} \leq 0 \tag{HC1-CBF}$$

where

$$h(x,z) = z - 1.8x_2 \tag{20}$$

$$B(x,z) = -log(\frac{h(x)}{1+h(x)}) \tag{21}$$

$$L_f B(x,z) = -\frac{1.8F_r(x) + m(v_0 - x_2)}{m(1 - 1.8x_2 + z)(-1.8x_2 + z)} \tag{22}$$

$$L_g B(x,z) = \frac{1.8}{m(1 - 1.8x_2 + z)(-1.8x_2 + z)} \tag{23}$$

Proof: HC1-CBF

By the Lie Derivative, one have

$$\dot{B}(x,z) = L_f B(x,z) + L_g B(x,z)u \tag{24}$$

We also impose the constraint on $\dot{B}$ to be

$$\dot{B} \leq \frac{\gamma}{B} \tag{25}$$

Substituting and rearrangement to get

$$L_f B(x,z) + L_g B(x,z)u - \frac{\gamma}{B(x,z)} \leq 0 \tag{26}$$

Constraint II: Soft Constraint as CLF

Achieve a desired speed when hard constraints are satisfied

$$v - v_d \to 0 \tag{27}$$

Expression:

$$\psi_0(x, z) + \psi_1(x, z)u \leq \delta_{ac} \tag{SC1-CLF}$$

where

$$\psi_0(x, z) = -\frac{2(x_2 - v_d)}{m} F_r(x) + \varepsilon(x_2 - v_d)^2 \tag{28}$$

$$\psi_1(x, z) = \frac{2(x_2 - v_d)}{m} \tag{29}$$

Recall that the tracking error is $y = x_2 - v_d$. Define a Lyapunov function as

$$V = y^2 = (x_2 - v_d)^2 \tag{30}$$

$$\tag{31}$$

Differentiating and rearrangements yield

$$\dot{V} = 2y\dot{y} = 2(x_2 - v_d)(-\frac{1}{m}F_r(x) + \frac{1}{m}u) \tag{32}$$

$$\dot{V} = \frac{-2(x_2 - v_d)}{m}F_r(x) + \frac{2(x_2 - v_d)}{m}u \tag{33}$$

Inserting the relaxed ES-CLF condition $\dot{V} + \epsilon V \leq \delta_{sc}$

$$\frac{-2(x_2 - v_d)}{m}F_r(x) + \frac{2(x_2 - v_d)}{m}u + \epsilon(x_2 - v_d)^2 \leq \delta_{sc} \tag{34}$$

Rearrangement to get the result desired.

Constraint III:Force Based Constraints as CBF

allowable wheel forces, typically much less than the peak forces that can be generated by the car.

Expression:

$$L_f B_F(x,z) + L_g B_F(x,z)u - \frac{1}{B_F(x,z)} \leq 0 \tag{35}$$

Where

$$B_F(x,z) = \frac{1}{h_F(x,z)} \tag{36}$$

$$h_F(x,z) = -1.8x_2 - \frac{1}{2}\frac{(v_0 - x_2)^2}{c_d g} + z \tag{37}$$

$$L_f B_F = -\frac{1}{h_F(x,z)^2}\left[\left(1.8 - \frac{v_0 - x_2}{c_d g}\right)\frac{F_r}{m} + (v_0 - x_2)\right] \tag{38}$$

$$L_g B_F = \frac{-1}{m h_F^2}(-1.8 + \frac{v_0 - x_2}{c_d g}) \tag{39}$$

Constraint IV:Control Constraint

This defines the physical limits of the vehicle. how much it can accelerate or brake.

$$u \leq c_a m g \tag{FC1}$$

$$-u \leq c_d m g \tag{FC2}$$

Formulations

$$u^*(x,z) = \underset{\mathbf{u}=\begin{bmatrix} u \\ \delta_{sc} \end{bmatrix} \in \mathbb{R}^2}{\arg\min} \quad \frac{1}{2}\mathbf{u}^T H_{\mathsf{acc}} \mathbf{u} + F_{\mathsf{acc}}^T \mathbf{u}$$

$$\text{s.t.}$$

$$A_{\mathsf{clf}}\mathbf{u} \leq b_{\mathsf{clf}} \qquad \text{(CLF)}$$

$$A_{\mathsf{cbf}}\mathbf{u} \leq b_{\mathsf{cbf}} \qquad \text{(CBF)}$$

$$A_{\mathsf{fcbf}}\mathbf{u} \leq b_{\mathsf{fcbf}} \qquad \text{(FCBF)}$$

$$A_{\mathsf{fc}}\mathbf{u} \leq b_{\mathsf{fc}} \qquad \text{(CC)}$$

$$A_{\mathsf{clf}} = \begin{bmatrix} \psi_1(x,z) & -1 \end{bmatrix},$$
$$b_{\mathsf{clf}} = -\psi_0(x,z).$$

$$A_{\mathsf{cbf}} = \begin{bmatrix} L_g B(x,z) & 0 \end{bmatrix},$$
$$b_{\mathsf{cbf}} = -L_f B(x,z) + \frac{\gamma}{B(x,z)}.$$

$$A_{\mathsf{cc}} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}, \qquad b_{\mathsf{cc}} = \begin{bmatrix} c_a mg \\ c_d mg \end{bmatrix}.$$

$$A_{\mathsf{fcbf}} = \begin{bmatrix} L_g B_F(x,z) & 0 \end{bmatrix},$$
$$b_{\mathsf{fcbf}} = -L_f B_F(x,z) + \frac{1}{B_F(x,z)}.$$

Section 4

1. Motivation
2. Introduction
3. ACC Problem Setup
4. **Experiment**
5. Conclusion

Environment Setup

- Environment: Google Colab
- Package: qpsolver, math, numpy

The paper tests two cases

- Case I: QP only solved using CLF and CBF
- Case II: QP has all four constraints, including CLF, CBF, FCBF, and CC

Under each case, we also explore the effect of p_{sc}, which is related to the penalty of the ES-CLF slack

$$\dot{V} + \epsilon V \leq \delta_{sc} \tag{40}$$

$$\frac{1}{2}\mathbf{u}^T H_{acc}\mathbf{u} = \frac{1}{2}\begin{bmatrix} u & \delta_{sc} \end{bmatrix} \cdot 2 \begin{bmatrix} \frac{1}{m^2} & 0 \\ 0 & p_{sc} \end{bmatrix} \cdot \begin{bmatrix} u \\ \delta_{sc} \end{bmatrix} = \frac{u^2}{m^2} + p_{sc}\delta_{sc}^2 \tag{41}$$

Parameters used

Parameter	Value	Units
g	9.81	$\frac{m}{s^2}$
m	1650	kg
f_0	0.1	$Newton$
f_1	5	$\frac{(Newton)(s)}{m}$
f_2	0.25	$\frac{(Newton)(s^2)}{m}$
v_d	24	$\frac{m}{s}$
v_0	13.89	$\frac{m}{s}$
ε	10	Unitless
γ	1	Unitless
c_a	0.3	Unitless
c_d	0.3	Unitless
p_{sc}	$1e^{-5}$	Unitless

Table: Table of parameters used in the simulation results.

Case I

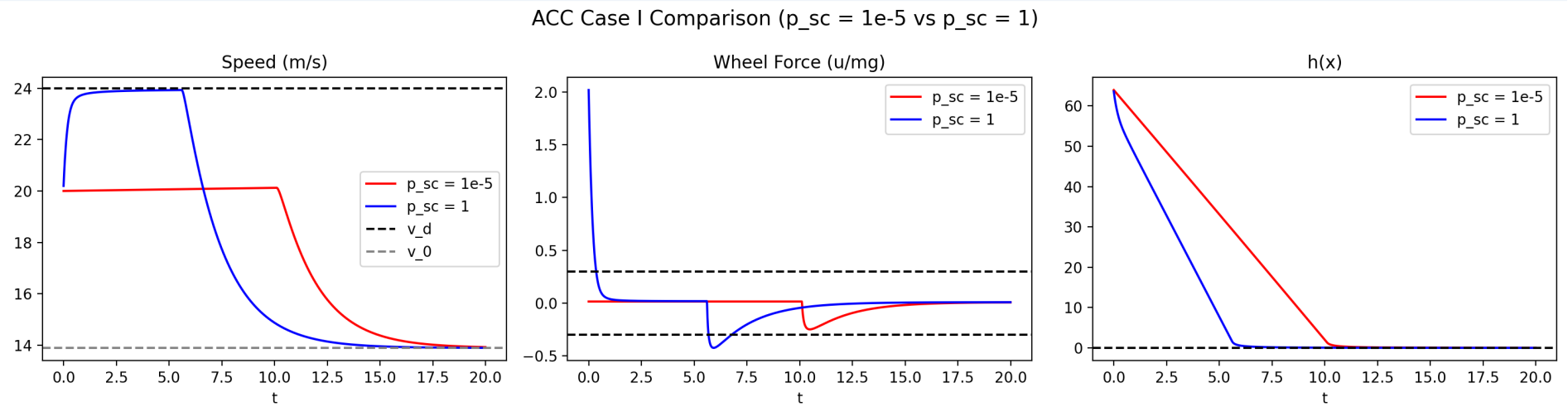

Figure: Case I simulation result

Case II

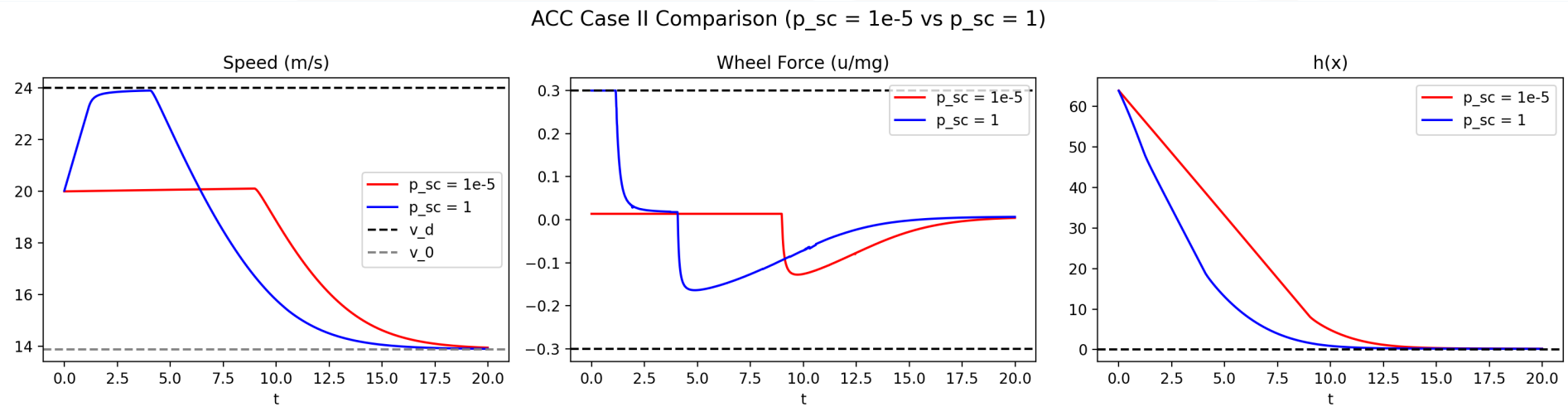

Figure: Case II simulation result

Section 5

1. Motivation
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3. ACC Problem Setup
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5. **Conclusion**

Conclusion and Future Work

Summary

- The author applied the CLF and CBF into adpative cruise control scenario; QP allows the integration of two concepts
- By including the full constraints (CLF, CBF, FCBF, CC), the ACC is able to perform braking without causing discomfor
- p_{sc} affects the behavior of system

Future Work

- Test with varying lead vehicle speed.
- Considering noisy sensor measurement with Kalman Filters
- Compare more rigourously with the PID and MPC baselines

References I

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[Pukdeboon, 2011] Chutiphon Pukdeboon (2011)
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Journal of Applied Sciences, 10(2), pp. 55–61.

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https://ctms.engin.umich.edu/CTMS/index.php?example=CruiseControl&section=ControlPID
Accessed: April 2026

Thank You!

Some concepts:

- Forward Invariance: $\mathcal{C}$ is forward-invariant if

$$x(0) \in \mathcal{C} \rightarrow x(t) \in \mathcal{C} \quad \forall t \geq 0 \tag{42}$$

- class κ function: A function $\alpha : [0, a) \rightarrow \mathbb{R}$ is a class κ if

$$\alpha(0) = 0, \alpha(r) > 0 \quad \forall r > 0. \text{ and } \alpha \text{ is strictly increasing.} \tag{43}$$

- Locally Lipschitz: $f : X \rightarrow Y$ is locally Lipschitz if for every $x \in X$, there exists a neighborhood U and a constant $L_U > 0$ such that

$$\|f(y_1) - f(y_2)\| \leq L_U \|y_1 - y_2\| \quad \forall y_1, y_2 \in U \tag{44}$$

  Intuition: It does not change too abruptly in any small region

General Notion of Barrier Functions

Definition 1: For the dynamical system (1), a function $B : \mathcal{C} \subset \mathbb{R}^n \to \mathbb{R}$ is a **barrier function (BF)** for the set $\mathcal{C}$ defined by (2)-(4) for a continuously differentiable function $h : \mathbb{R}^n \to \mathbb{R}$ if there exist locally Lipschitz class $\mathcal{K}$ functions $\alpha_1, \alpha_2, \alpha_3$ such that, for all $x \in \mathsf{Int}(\mathcal{C})$,

$$\frac{1}{\alpha_1(h(x))} \leq B(x) \leq \frac{1}{\alpha_2(h(x))} \tag{45}$$

$$\dot{B}(x) \leq \alpha_3(h(x)). \tag{46}$$

The motivation for the conditions on barrier functions is the following: Condition (7) implies that the barrier function B essentially behaves like the function $\frac{1}{\alpha(h(x))}$ for α a class $\mathcal{K}$ function which satisfies the essential conditions:

$$\inf_{x \in \mathsf{Int}(\mathcal{C})} \frac{1}{\alpha(h(x))} \geq 0, \qquad \lim_{x \to \partial\mathcal{C}} \frac{1}{\alpha(h(x))} = \infty.$$

Moreover, the condition on $\dot{B}$ in (8) allows for B to grow quickly when solutions are far away from $\partial\mathcal{C}$, with this growth approaching zero as solutions approach $\partial\mathcal{C}$.

Proof: $L_g B_F$

Construct an augmented state $\bar{x} = (x_1, x_2, z)$, and then

$$\dot{\bar{x}} = \begin{bmatrix} \dot{x_1} \\ \dot{x_2} \\ \dot{z} \end{bmatrix} = \underbrace{\begin{bmatrix} x_2 \\ \frac{-1}{m} F_r \\ v_0 - x_2 \end{bmatrix}}_{f(x,z)} + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{m} \\ 0 \end{bmatrix}}_{g(x,z)} u \tag{47}$$

Proof: $L_g B_F$

$$L_g B_F = \nabla B_F \cdot g \tag{48}$$

$$= \frac{\partial B_F}{\partial h_F} \nabla h_F \cdot g \tag{49}$$

$$= \left(-\frac{1}{h_F^2} \right) L_g h_F \tag{50}$$

$$\tag{51}$$

We now compute $L_g h_F$. Known that

$$h_F(x,z) = z - 1.8x_2 - \frac{(v_0 - x_2)^2}{2c_d g} \tag{52}$$

$$\begin{aligned} L_g h_F &= \nabla h_F \cdot g \\ &= \begin{bmatrix} 0 \\ -1.8 + \frac{v_0 - x_2}{c_d g} \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ \frac{1}{m} \\ 0 \end{bmatrix} \\ &= \frac{1}{m}(-1.8 + \frac{v_0 - x_2}{c_d g}) \end{aligned} \tag{53}$$

We compute $L_g B_F$ together from (23) and (26)

$$L_g B_F = \frac{-1}{m h_F^2}(-1.8 + \frac{v_0 - x_2}{c_d g}) \tag{54}$$